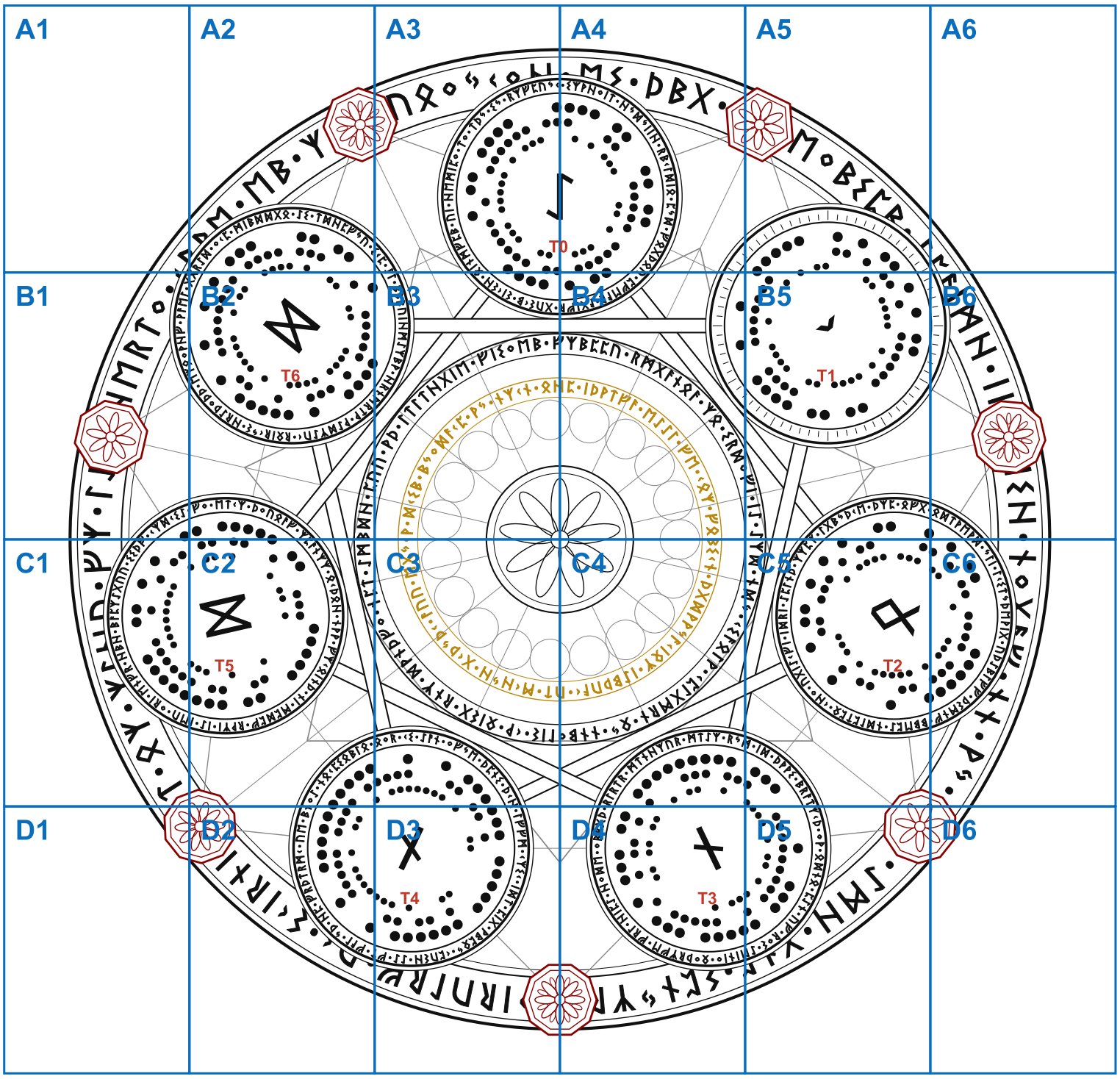
Magic circle — A3 assembly kit

seed 576808920 · finished circle Ø 140.6 cm · 24 poster sheets + 7 tag patches = 33 pages



Before printing

- Paper size A3, scale 100 % / "Actual size". Turn OFF "Fit to page", "Shrink oversized pages" and borderless printing.
- Every page has a 100 mm ruler. Measure it on the first printed page; it must be 100 mm (±0.5 mm), otherwise fix the settings.
- Prefer matte paper and a laser printer: gloss causes glare that hides the tag dots from cameras.

Key dimensions

- RUNE-Tag dot field: outer edge of the outermost dots Ø 270.0 mm (as requested).
- RUNE-Tag model for pose estimation: outer dot layer centre radius 127.82 mm; layers at 4/6, 5/6, 6/6 of it; dot radius = layer radius / 17.8.
- Tag centres sit on a circle of radius 490.5 mm around the circle centre (see page 2 for coordinates).
- One generator pixel = 0.7280 mm. Grid: 6 columns × 4 rows of 265 × 382 mm visible area + 12 mm glue strips.



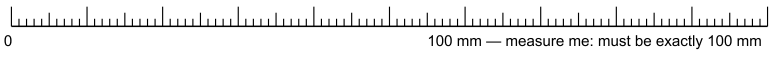
Assembly

1. Lay the sheets out as in the map on page 1 (rows A, B, … top to bottom; columns 1, 2, … left to right).
2. Trim: on each sheet cut ONLY along the black dashed "CUT" lines (left and/or top edge). Keep the blue glue strips.
3. Build each row from left to right: lay the next sheet over the blue glue strip of its left neighbour so that its cut edge lies exactly on the blue edge line. Every line of the artwork must continue straight across the seam. Fix with glue stick or double-sided tape on the blue strip only.
4. Join the rows from top to bottom the same way, using the blue strips along the bottom of the upper row.
5. Tag patches (last 7 pages): the dot rings would otherwise be split by seams. Cut each disc out along its grey circle and glue it over the matching tag (T0…T6) on the poster. The short ticks on the disc edge must meet the ticks on the poster; the DOUBLE tick points to the centre of the magic circle. Never rotate a patch.
6. Press everything flat: bubbles and wrinkles bend the dot rings. Mount on a flat, rigid, matte surface.

Tags (positions in mm from the circle centre; x to the right, y up)

Tag	Clock position	RUNE-Tag IDX	x (mm)	y (mm)	Patch page
T0	12.00 o'clock	8	0.0	490.5	27
T1	1.71 o'clock	10	383.5	305.8	28
T2	3.43 o'clock	15	478.2	-109.2	29
T3	5.14 o'clock	23	212.8	-442.0	30
T4	6.86 o'clock	24	-212.8	-442.0	31
T5	8.57 o'clock	26	-478.2	-109.2	32
T6	10.29 o'clock	29	-383.5	305.8	33

The "RUNE-Tag IDX" is the marker index in the official RUNE-Tag codes.txt; T0…T6 is the order on the circle (clockwise from the top).
The finished size depends on accurate printing and assembly: check the rulers, and measure one tag after gluing (dots Ø 270 mm).



A1

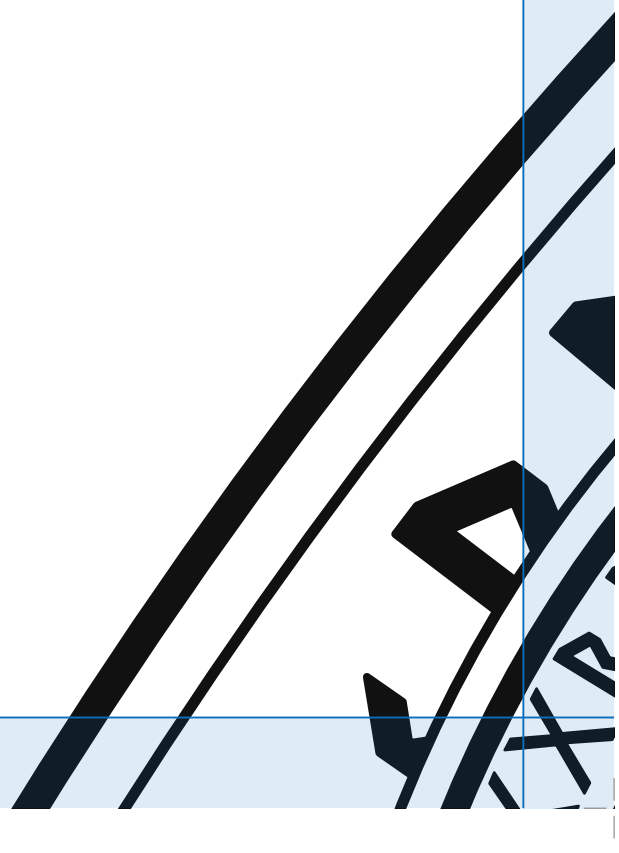
row A of D · column 1 of 6 · seed 576808920 · A3 @ 100 %

0 100 mm — measure me: must be exactly 100 mm

GLUE · A2 goes on top

A2 →

GLUE · B1 goes on top



↓ B1

A2

row A of D · column 2 of 6 · seed 576808920 · A3 @ 100 %

0 100 mm — measure me: must be exactly 100 mm

CUT



GLUE · A3 goes on top

GLUE · B2 goes on top

← A1

A3 →

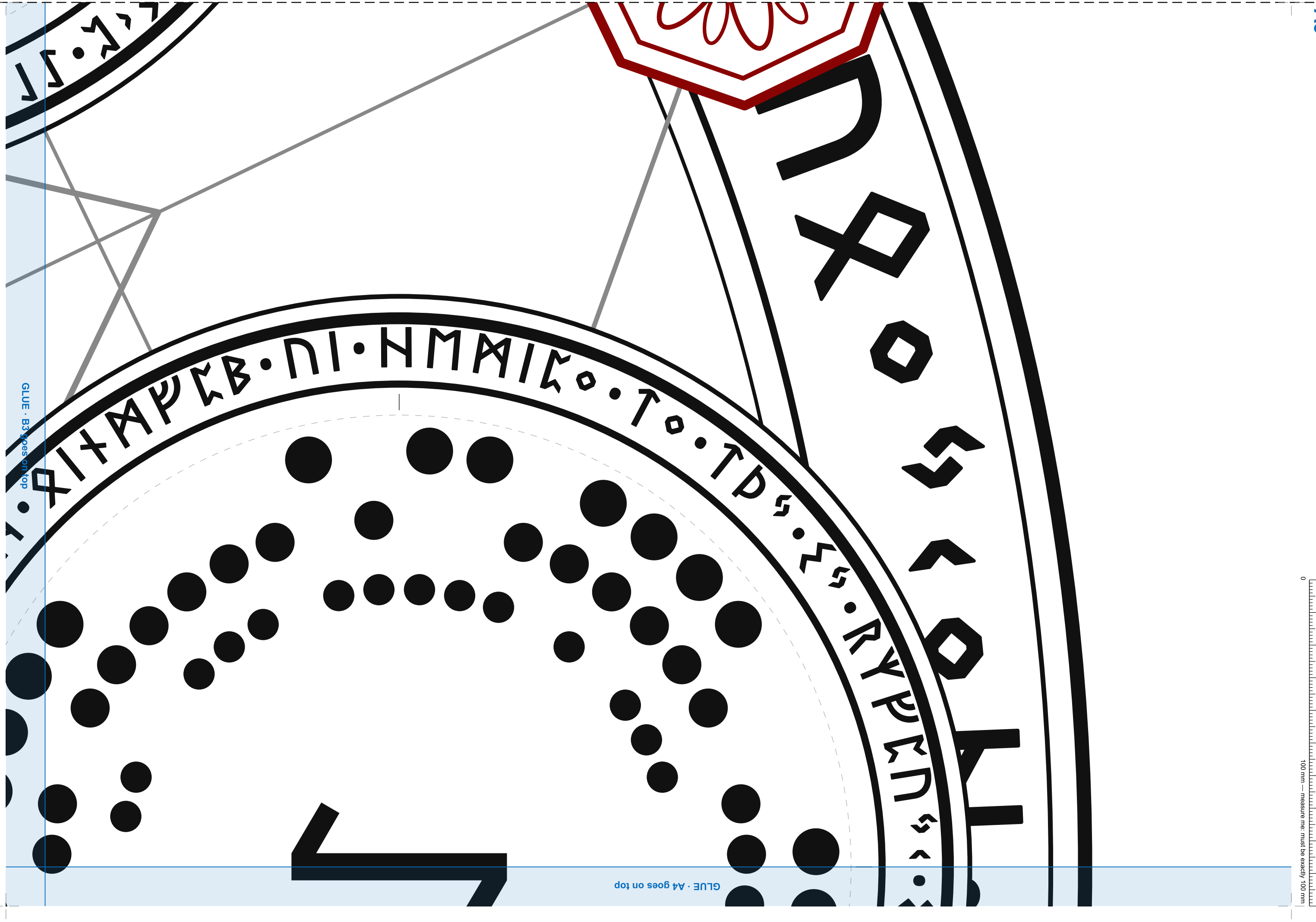
↓ B2

← A2

CUT

A3

row A of D · column 3 of 6 · seed 576808920 · A3 @ 100 %



GLUE · B3 goes on top

↑ B3

GLUE · A4 goes on top

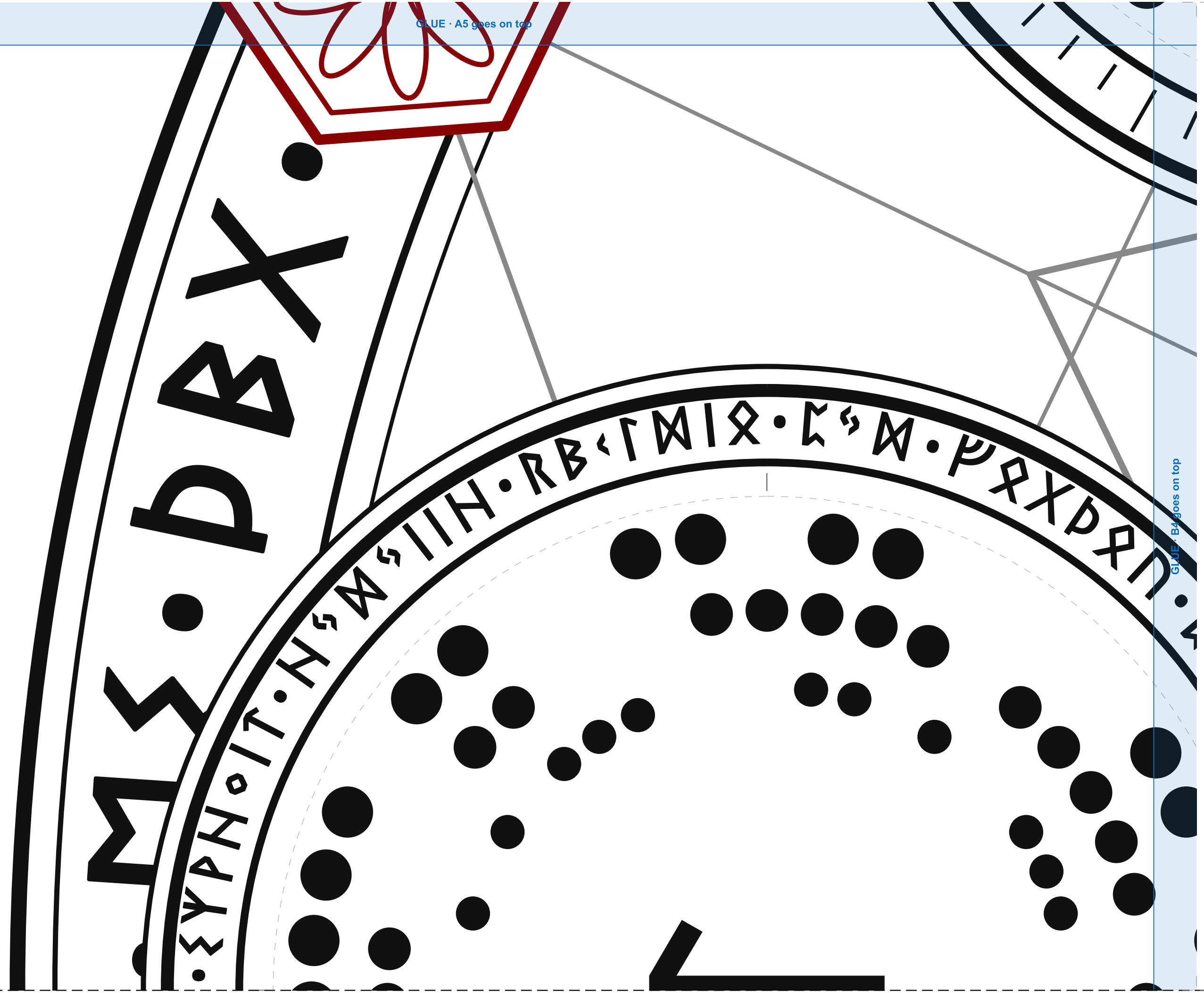
A4 →

0

100 mm — measure me: must be exactly 100 mm

A5 →

GLUE · A5 goes on top



GLUE · B4 goes on top

↓ B4

100 mm — measure me: must be exactly 100 mm

0

A4

row A of D · column 4 of 6 · seed 576808920 · A3 @ 100 %

CUT

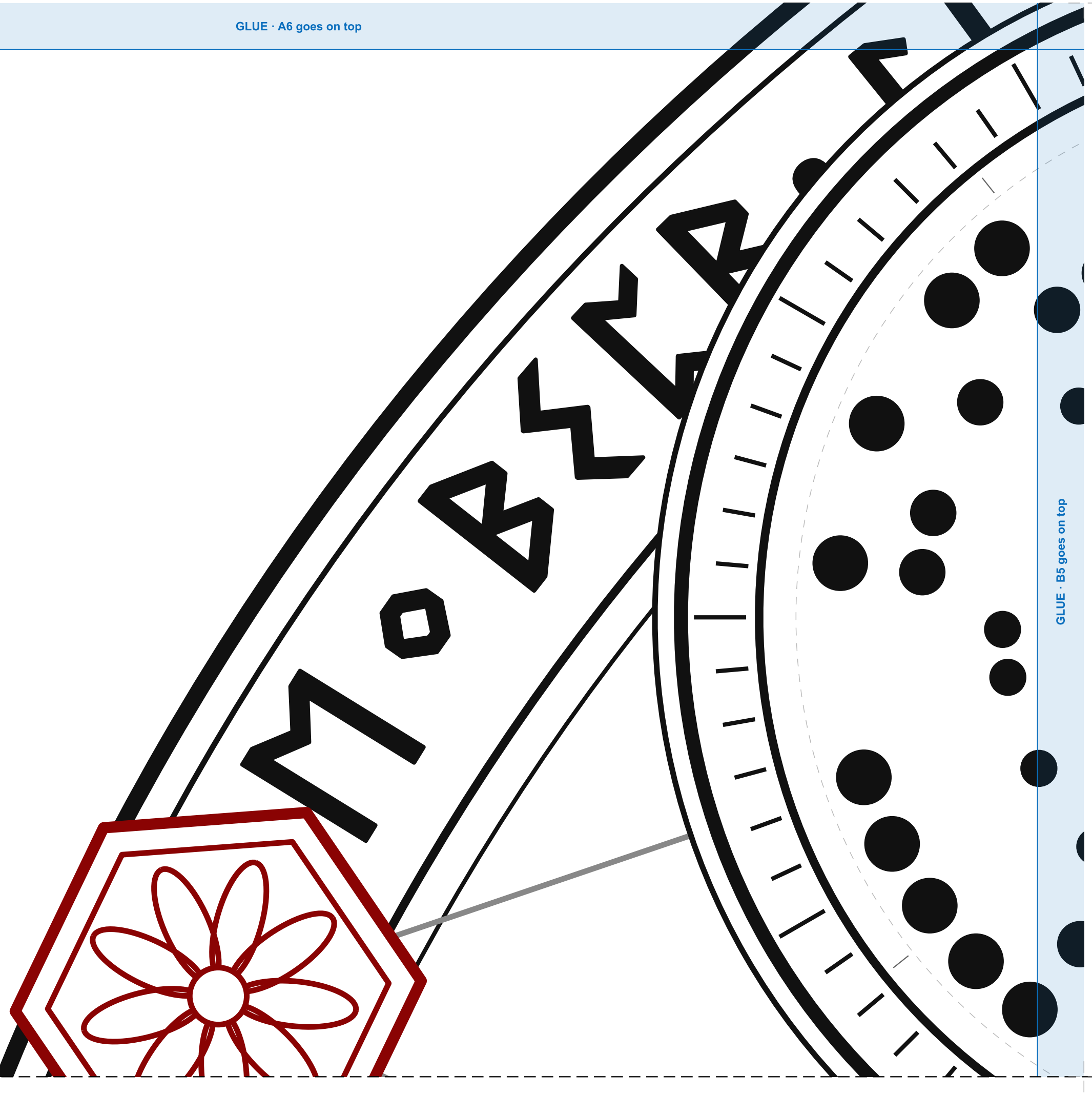
← A3

A6 →

0

100 mm — measure me: must be exactly 100 mm

GLUE · A6 goes on top



GLUE · B5 goes on top

↓ B5

A5

row A of D · column 5 of 6 · seed 576808920 · A3 @ 100 %

CUT

← A4

A6

row A of D · column 6 of 6 · seed 576808920 · A3 @ 100 %

0 100 mm — measure me: must be exactly 100 mm

CUT

← A5

GLUE · B6 goes on top

↓ B6

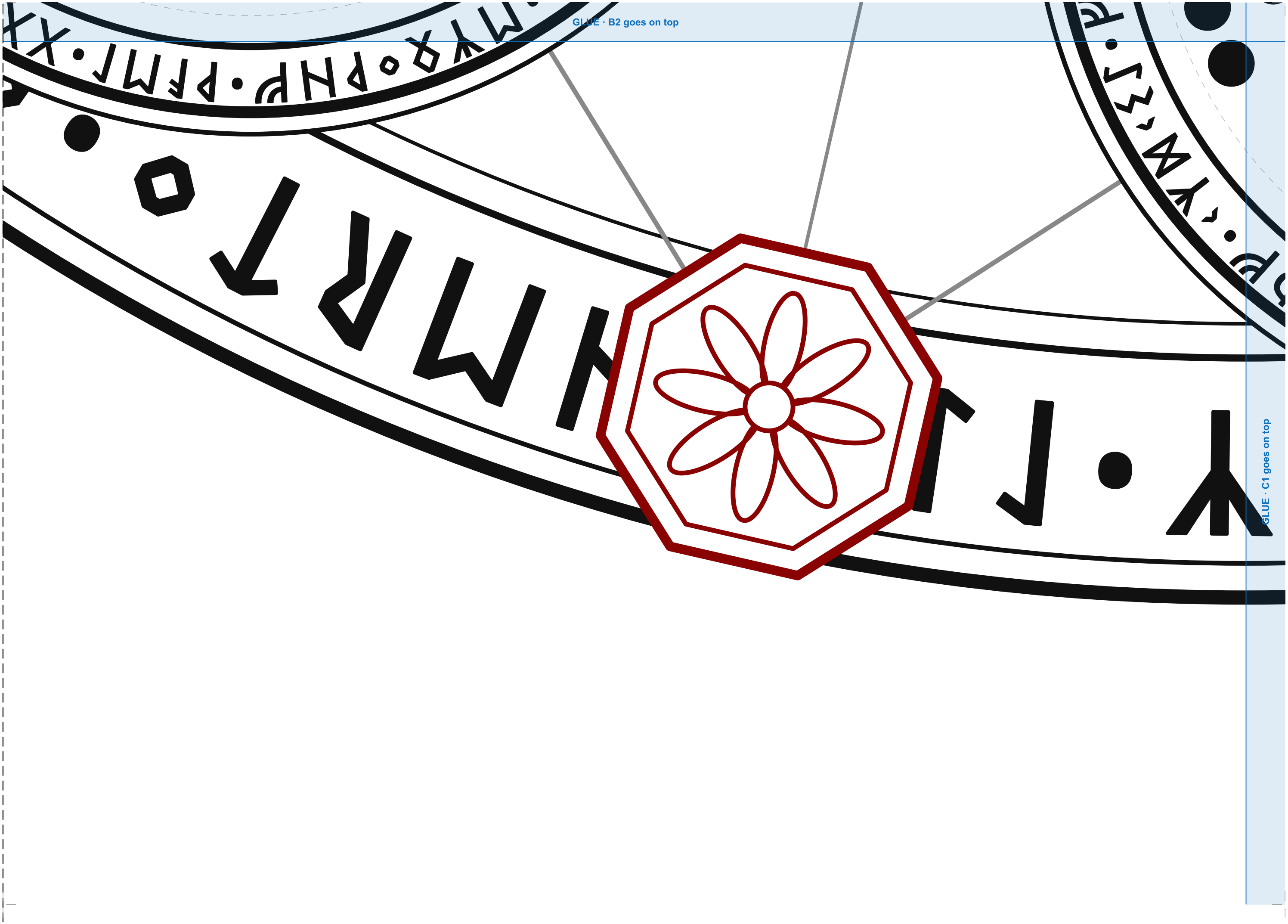
B2 →

GLUE · B2 goes on top

100 mm — measure me: must be exactly 100 mm

0

CUT



GLUE · C1 goes on top

↓ C1

B1

row B of D · column 1 of 6 · seed 576808920 · A3 @ 100 %

B2

row B of D · column 2 of 6 · seed 576808920 · A3 @ 100 %

0 100 mm — measure me: must be exactly 100 mm

CUT

CUT

GLUE · B3 goes on top

← B1

B3 →

GLUE · C2 goes on top

↓ C2

B4 →

GLUE · B4 goes on top

100 mm — measure me: must be exactly 100 mm

0

CUT

GLUE · C3 goes on top

↓ C3

B3

row B of D · column 3 of 6 · seed 576808920 · A3 @ 100 %

CUT

← B2

B4

row B of D · column 4 of 6 · seed 576808920 · A3 @ 100 %

0 100 mm — measure me: must be exactly 100 mm

CUT

CUT

GLUE · B5 goes on top

← B3

B5 →

GLUE · C4 goes on top

↓ C4

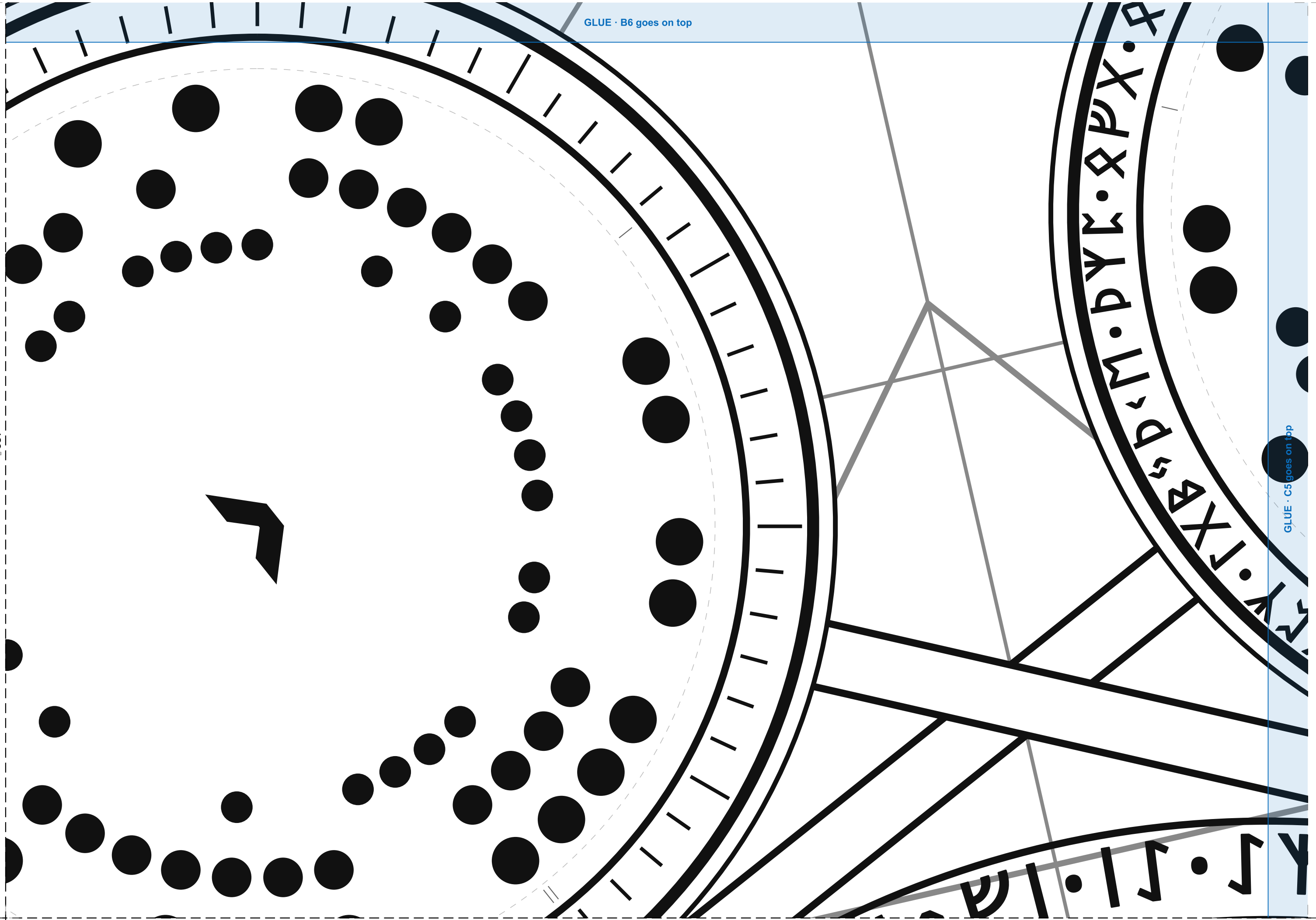
B6 →

100 mm — measure me: must be exactly 100 mm

0

CUT

GLUE · B6 goes on top



GLUE · C5 goes on top

↓ C5

B5

row B of D · column 5 of 6 · seed 576808920 · A3 @ 100 %

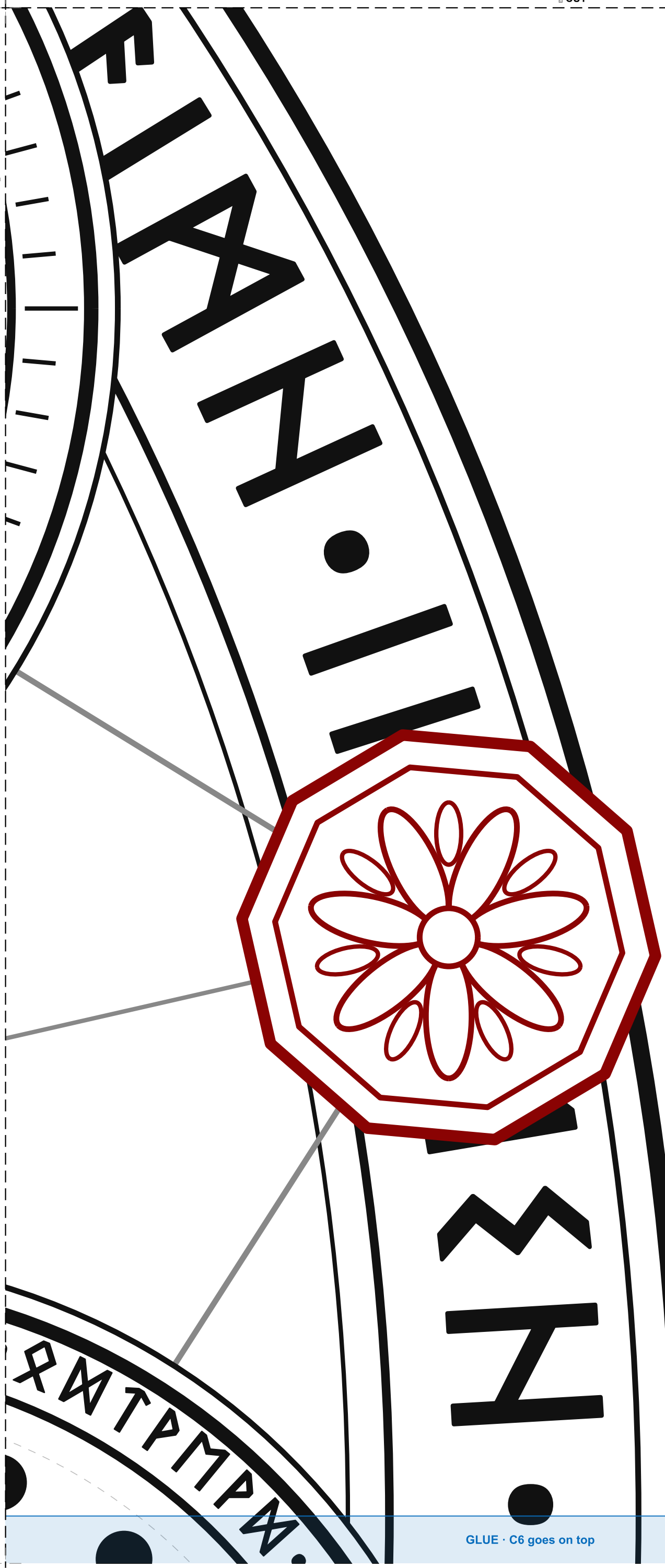
CUT

← B4

0 100 mm — measure me: must be exactly 100 mm

CUT

CUT



← B5

GLUE · C6 goes on top

↓ C6

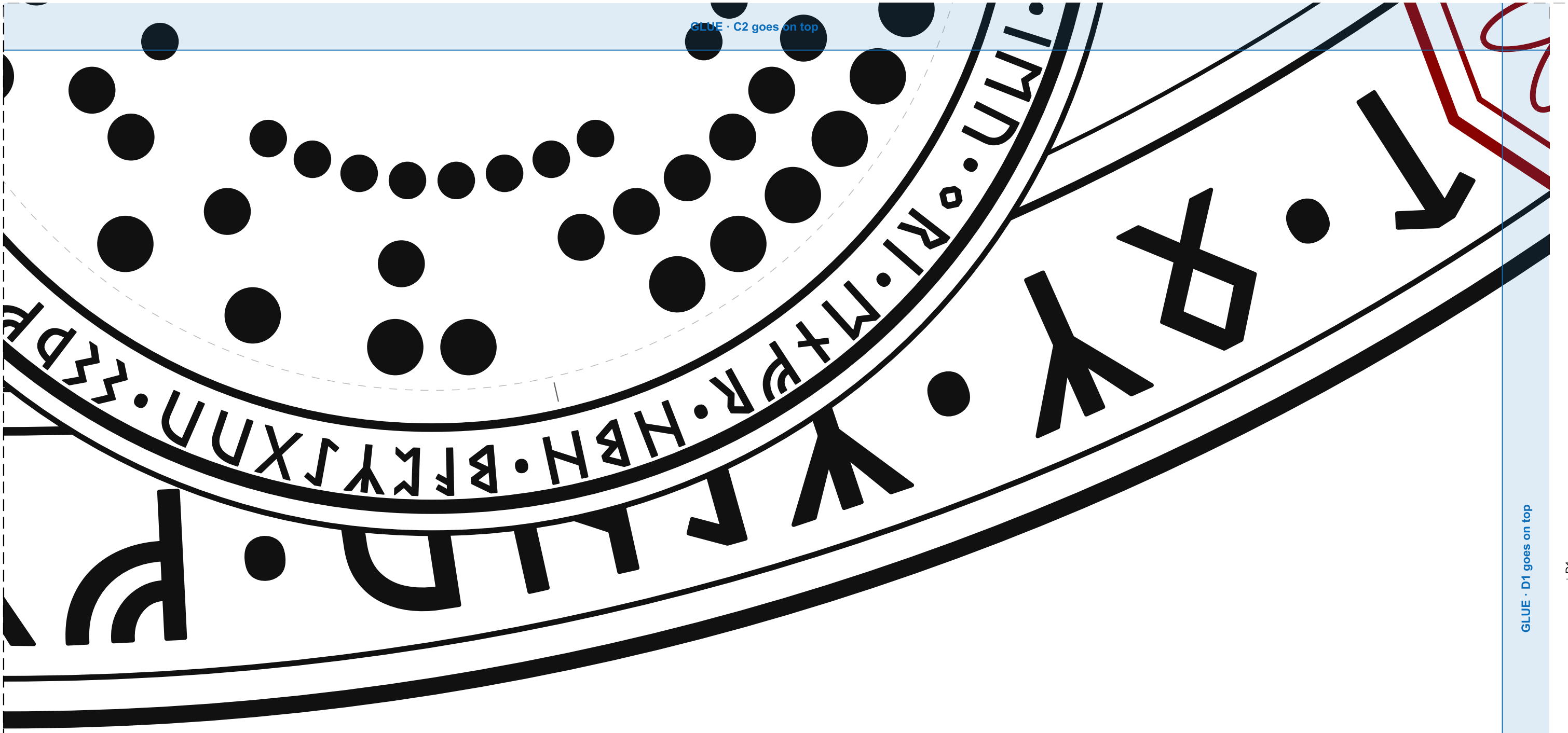
C2 →

GLUE · C2 goes on top

100 mm — measure me: must be exactly 100 mm

0

CUT



GLUE · D1 goes on top

↓ D1

C1

row C of D · column 1 of 6 · seed 576808920 · A3 @ 100 %

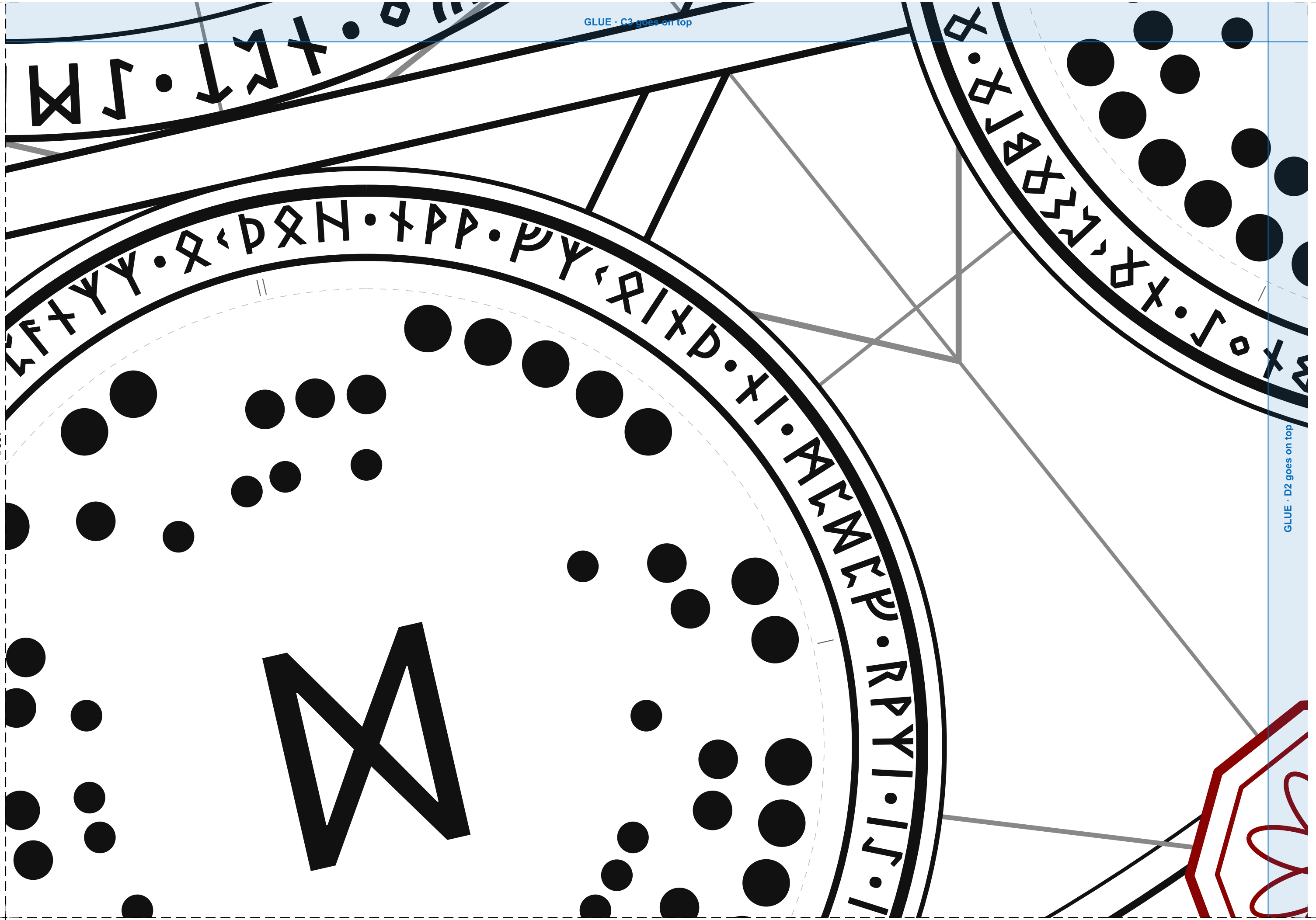
C3 →

GLUE · C3 goes on top

100 mm — measure me: must be exactly 100 mm

0

CUT



GLUE · D2 goes on top

↓ D2

C2

row C of D · column 2 of 6 · seed 576808920 · A3 @ 100 %

CUT

← C1

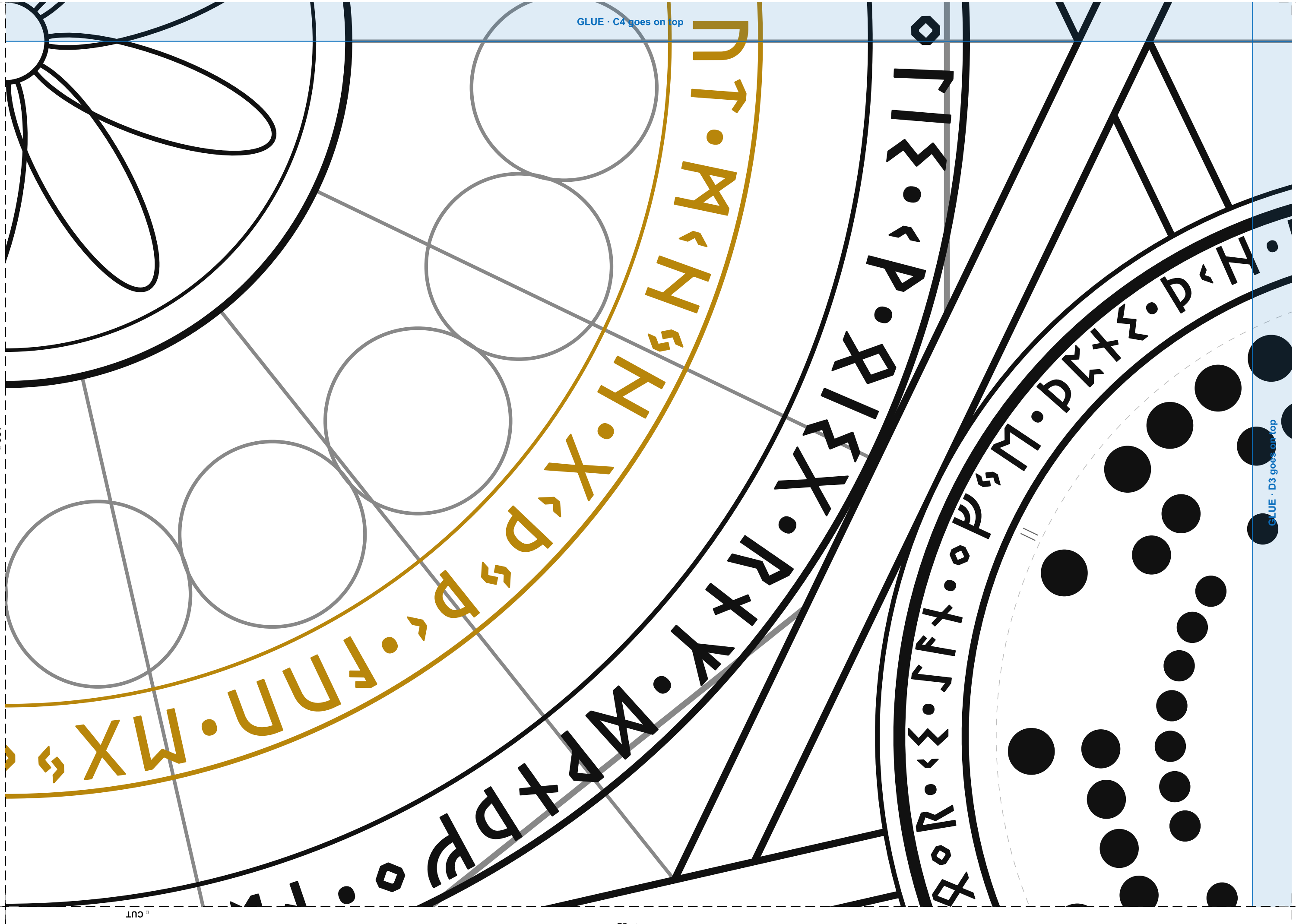
C4 →

GLUE · C4 goes on top

100 mm — measure me: must be exactly 100 mm

0

CUT



GLUE · D3 goes on top

↓ D3

C3

row C of D · column 3 of 6 · seed 576808920 · A3 @ 100 %

CUT

← C2

C4

row C of D · column 4 of 6 · seed 576808920 · A3 @ 100 %

0

100 mm — measure me: must be exactly 100 mm

CUT

CUT



GLUE · C5 goes on top

← C3

C5 →

GLUE · D4 goes on top

↓ D4

C6 →

GLUE · C6 goes on top

100 mm — measure me: must be exactly 100 mm

0

CUT

GLUE · D5 goes on top

↓ D5

C5

row C of D · column 5 of 6 · seed 576808920 · A3 @ 100 %

CUT

← C4

0
100 mm — measure me: must be exactly 100 mm

CUT

GLUE · D6 goes on top

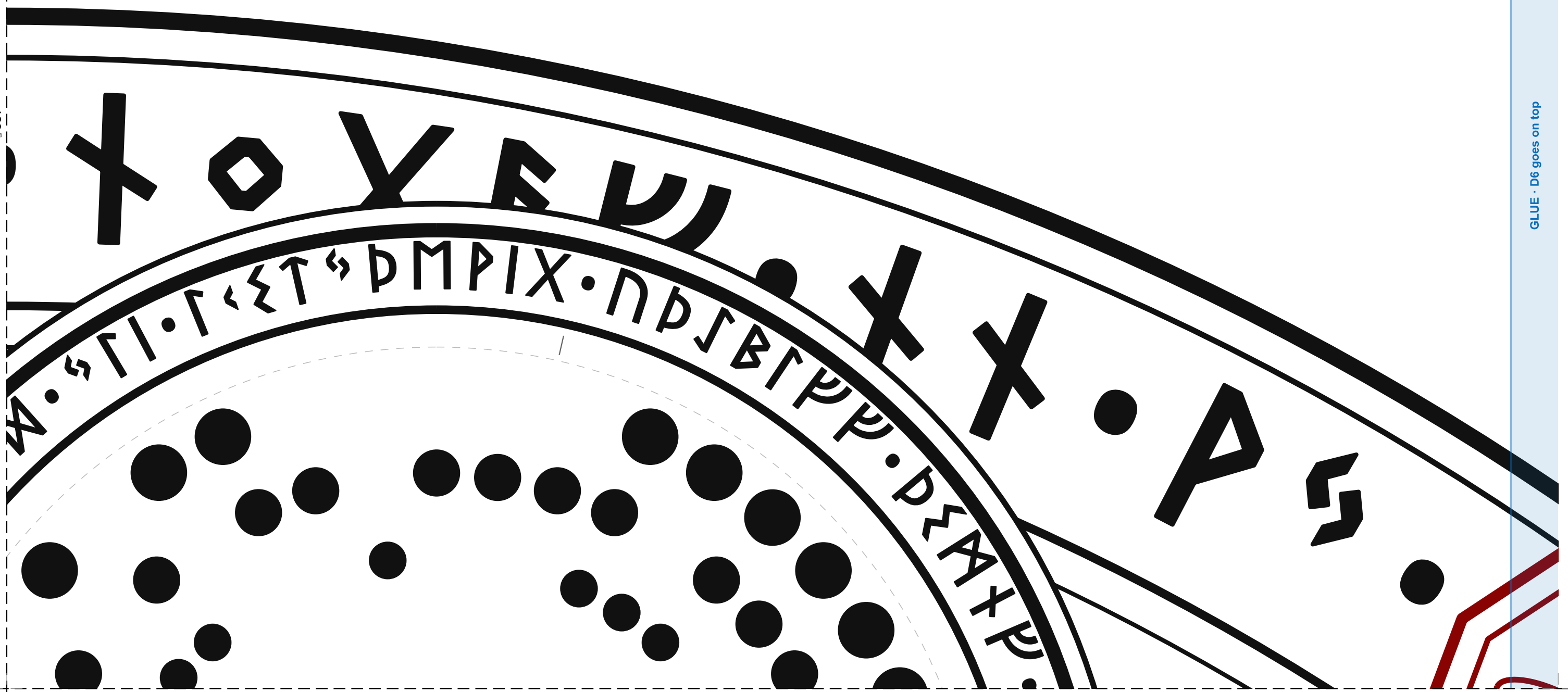
D6 →

← C5

CUT

C6

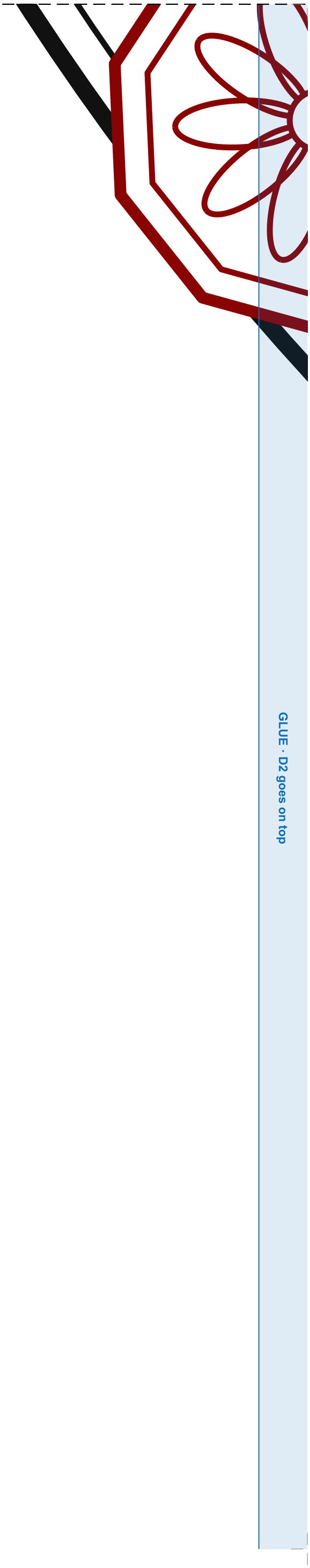
row C of D · column 6 of 6 · seed 576808920 · A3 @ 100 %



CUT

0

100 mm — measure me: must be exactly 100 mm



GLUE · D2 goes on top

D2 →

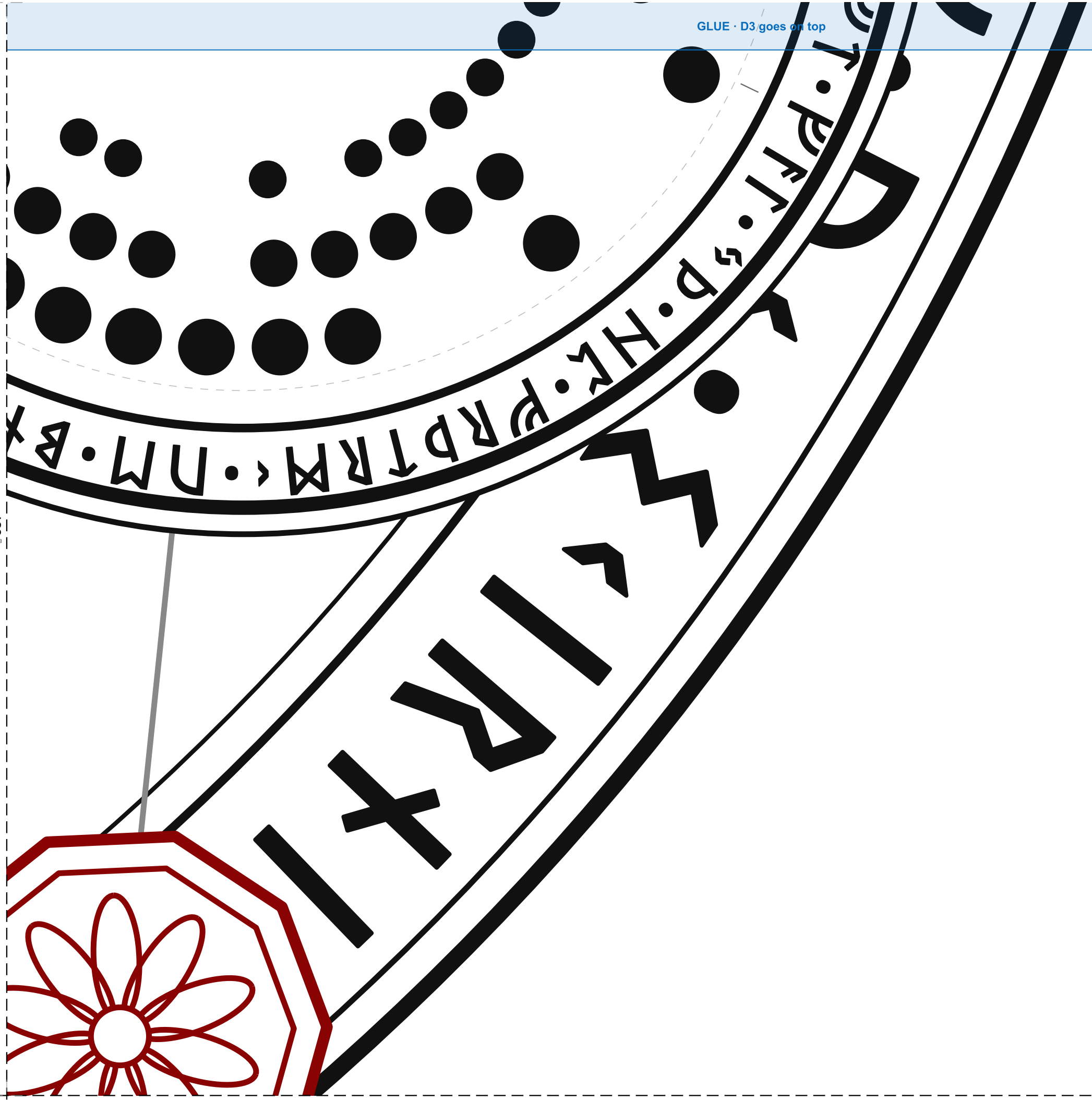
D3 →

GLUE · D3 goes on top

100 mm — measure me: must be exactly 100 mm

0

CUT



CUT

D2

row D of D · column 2 of 6 · seed 576808920 · A3 @ 100 %

← D1

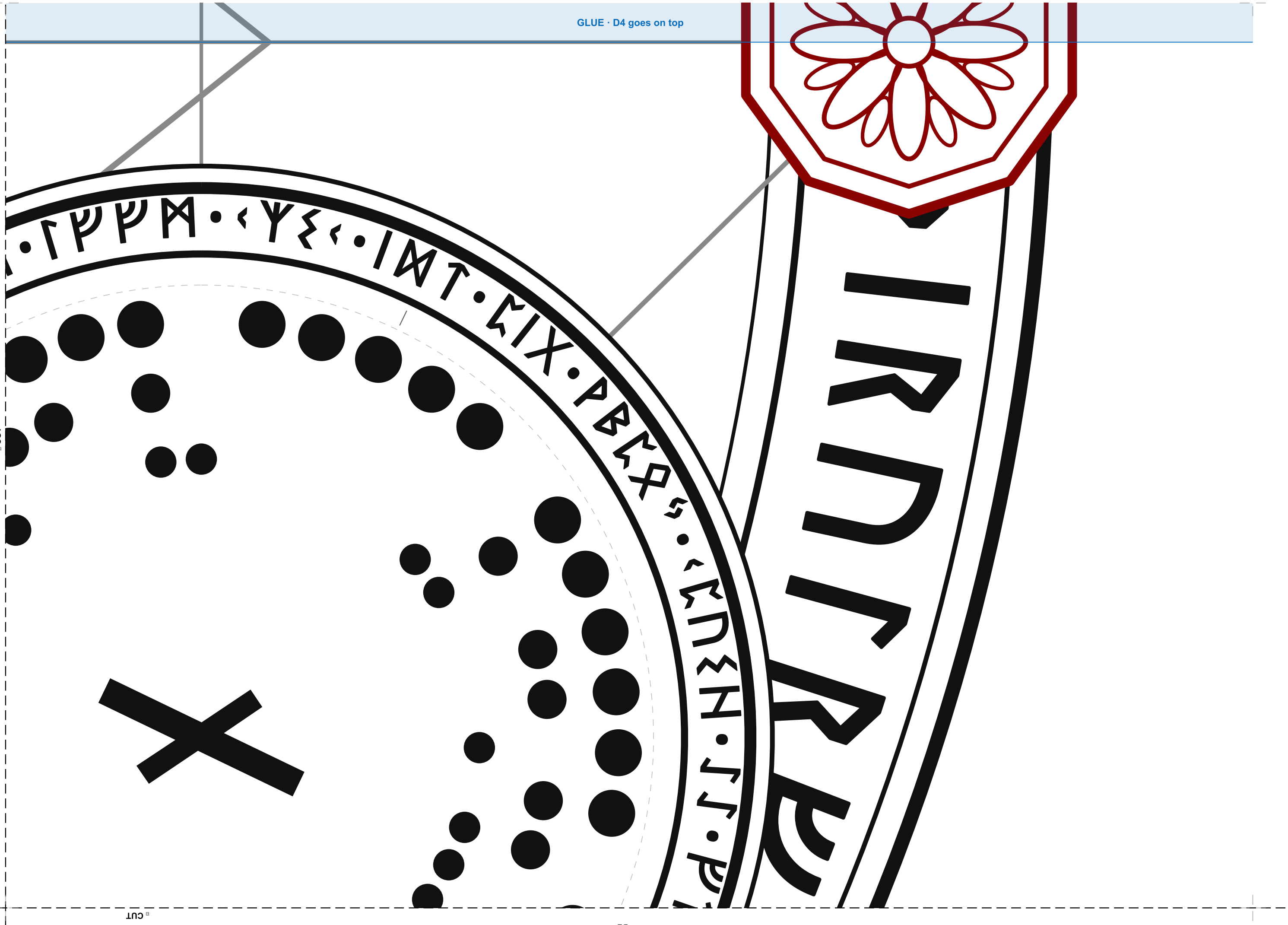
D4 →

GLUE · D4 goes on top

100 mm — measure me: must be exactly 100 mm

0

CUT



D3

row D of D · column 3 of 6 · seed 576808920 · A3 @ 100 %

CUT

← D2

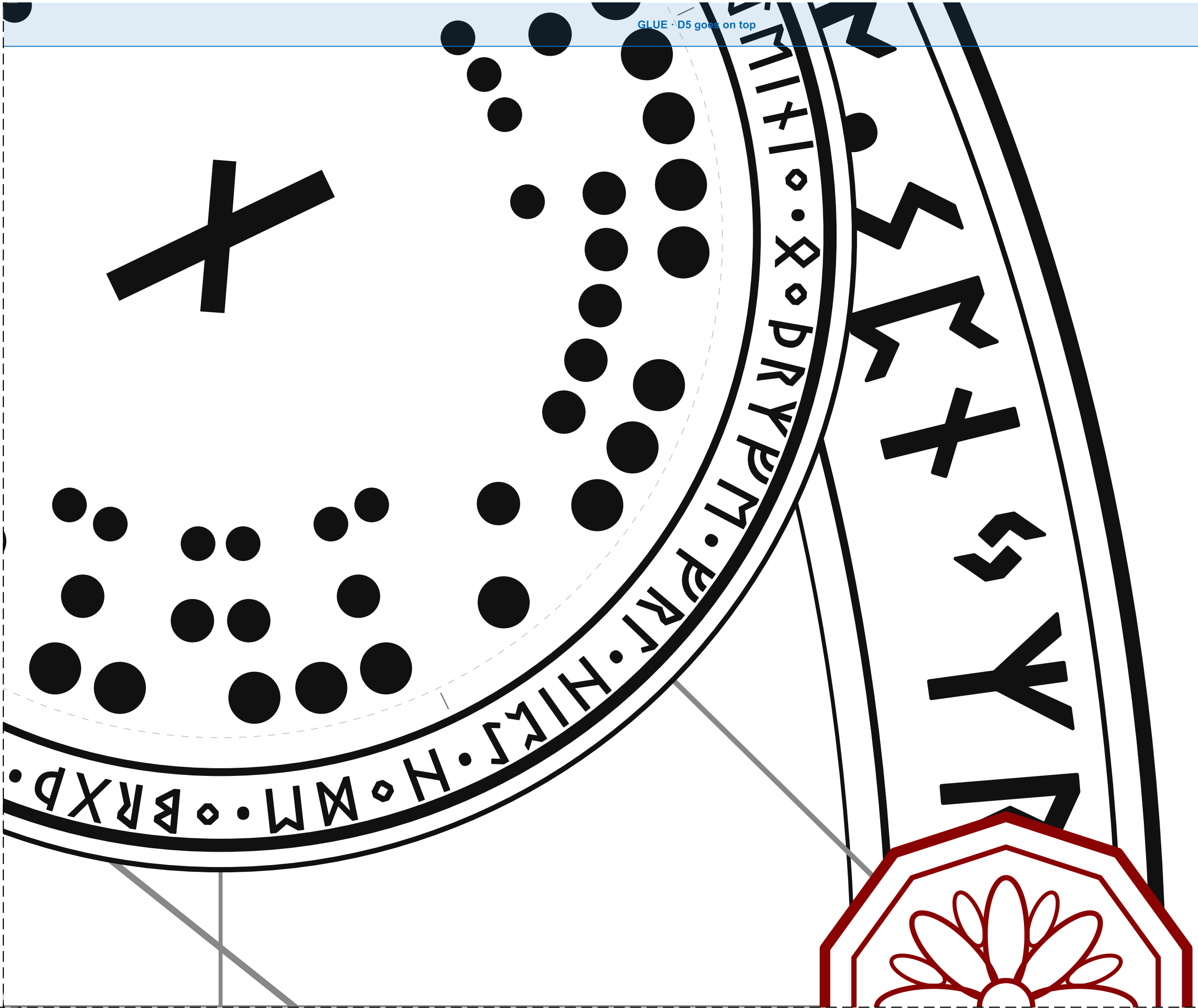
D5 →

GLUE · D5 goes on top

100 mm — measure me: must be exactly 100 mm

0

CUT



CUT

D4

row D of D · column 4 of 6 · seed 576808920 · A3 @ 100 %

← D3

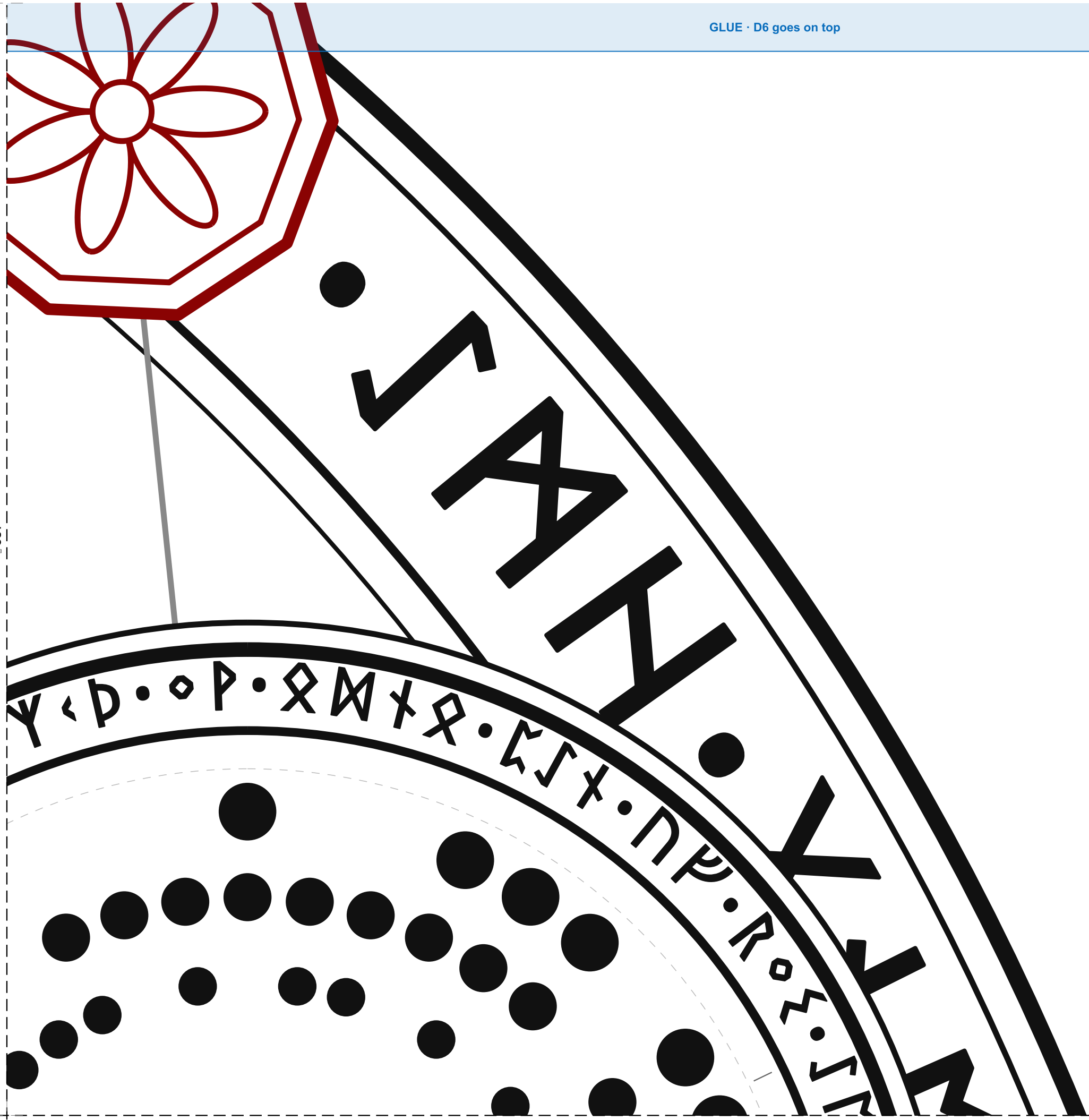
D6 →

GLUE · D6 goes on top

100 mm — measure me: must be exactly 100 mm

0

CUT



D5

row D of D · column 5 of 6 · seed 576808920 · A3 @ 100 %

CUT

← D4

D6

row D of D · column 6 of 6 · seed 576808920 · A3 @ 100 %

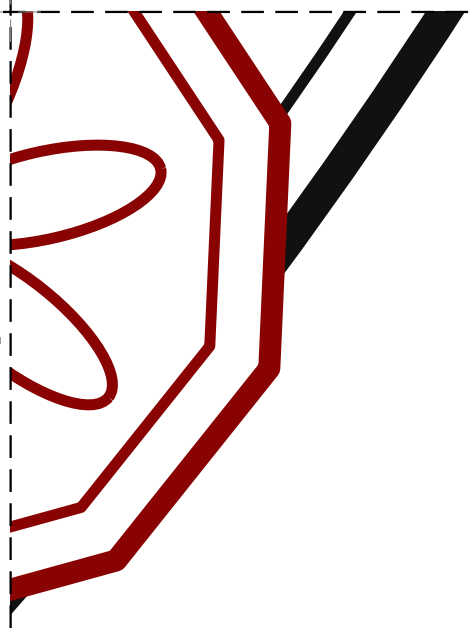
CUT

CUT

0

100 mm — measure me: must be exactly 100 mm

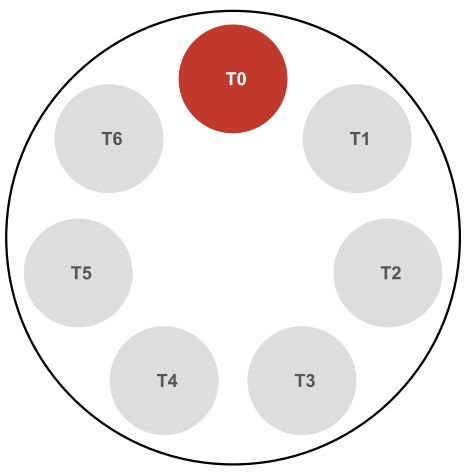
← D5



T0 patch · RUNE-Tag IDX 8 · seed 576808920

Cut along the grey circle. Double tick → circle centre.

▲ to circle centre



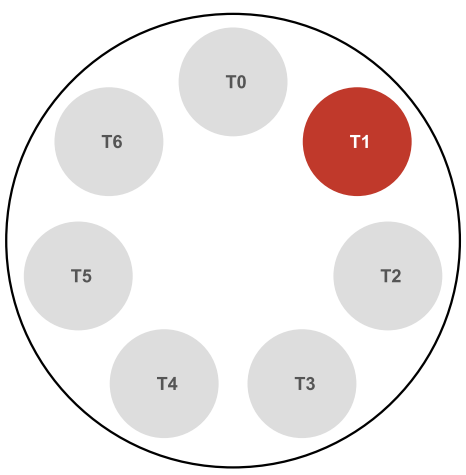
Glue this disc over tag T0 on the assembled poster
(the tag at 12.00 o'clock, top).
Line up all four ticks with the ticks on the poster.
Spread glue thinly and evenly; press flat from the centre out.
Dots: outer edge Ø 270 mm. Disc: Ø 280 mm.

0 — 100 mm — measure me: must be exactly 100 mm

T1 patch · RUNE-Tag IDX 10 · seed 576808920

Cut along the grey circle. Double tick → circle centre.

▲ to circle centre

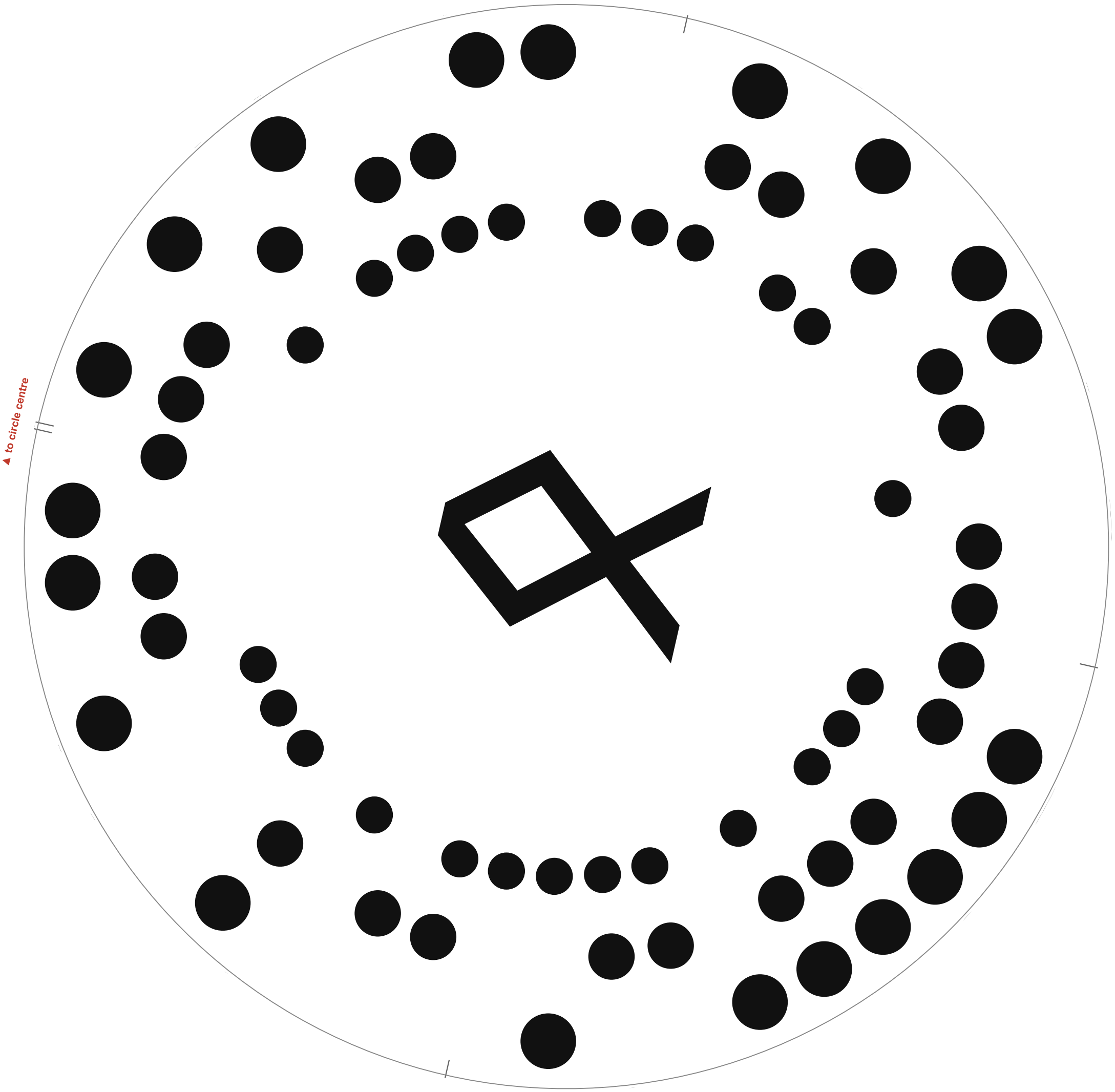


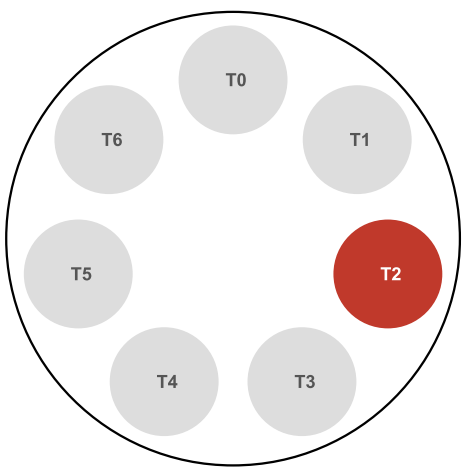
Glue this disc over tag T1 on the assembled poster
(the tag at 1.71 o'clock, upper right).
Line up all four ticks with the ticks on the poster.
Spread glue thinly and evenly; press flat from the centre out.
Dots: outer edge Ø 270 mm. Disc: Ø 280 mm.

0 — 100 mm — measure me: must be exactly 100 mm

T2 patch · RUNE-Tag IDX 15 · seed 576808920

Cut along the grey circle. Double tick → circle centre.



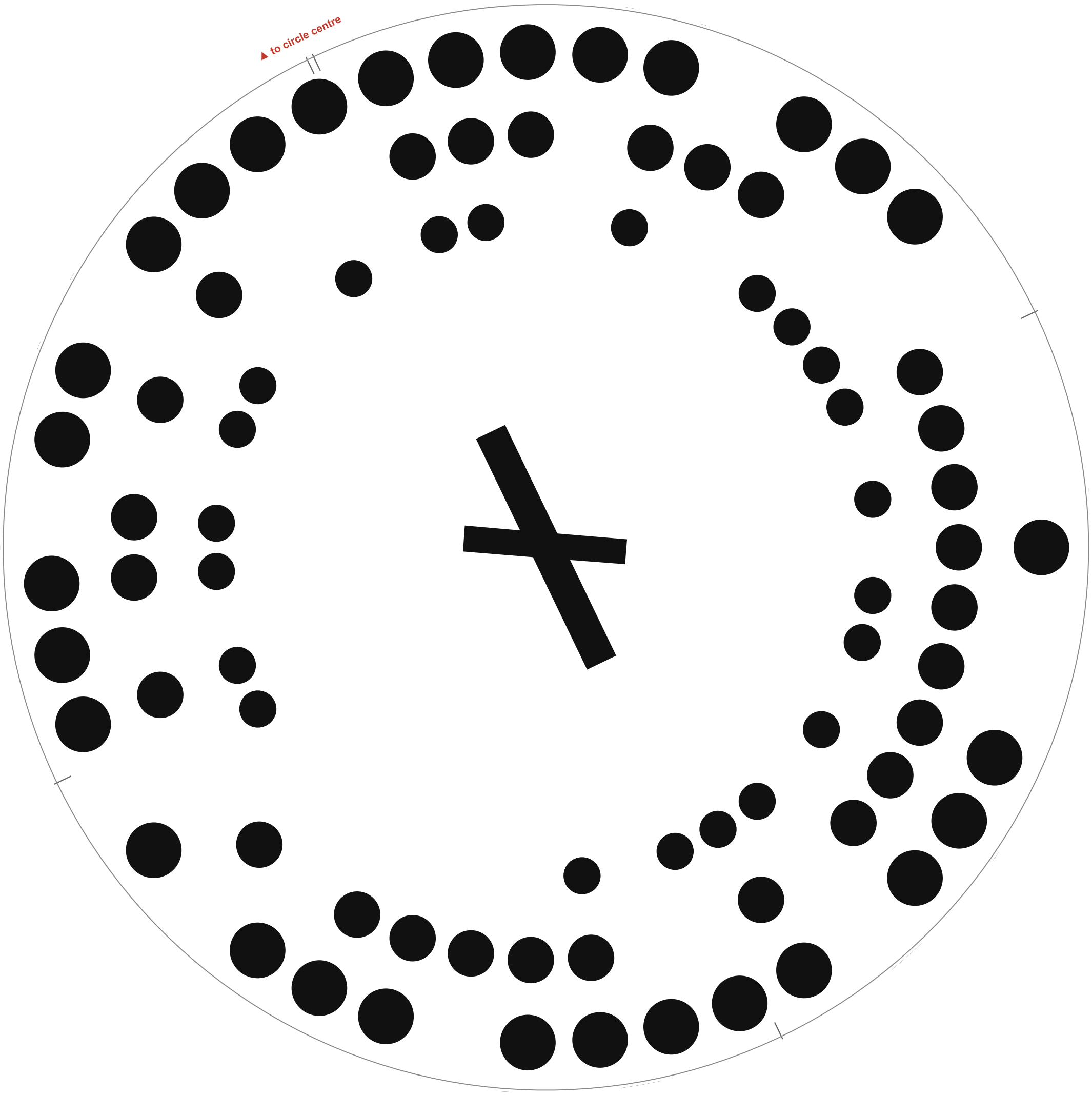


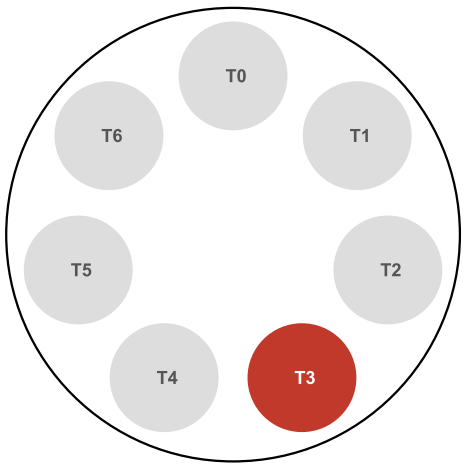
Glue this disc over tag T2 on the assembled poster
(the tag at 3.43 o'clock, right).
Line up all four ticks with the ticks on the poster.
Spread glue thinly and evenly; press flat from the centre out.
Dots: outer edge Ø 270 mm. Disc: Ø 280 mm.



T3 patch · RUNE-Tag IDX 23 · seed 576808920

Cut along the grey circle. Double tick → circle centre.



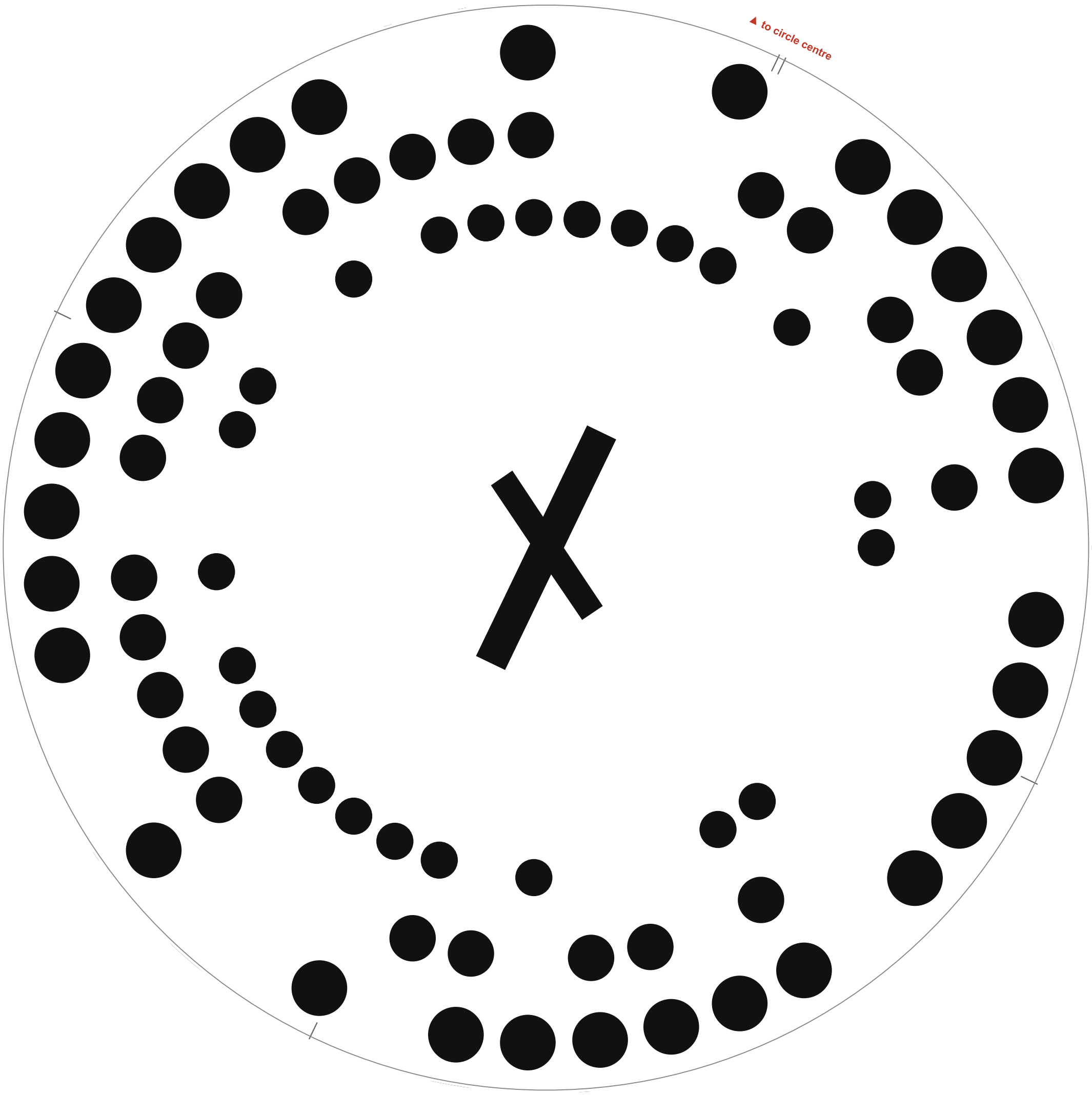


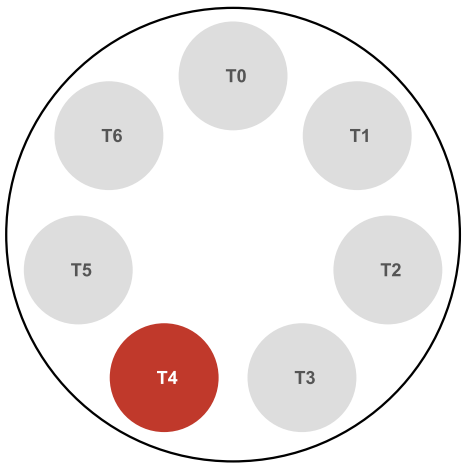
Glue this disc over tag T3 on the assembled poster
(the tag at 5.14 o'clock, lower right).
Line up all four ticks with the ticks on the poster.
Spread glue thinly and evenly; press flat from the centre out.
Dots: outer edge Ø 270 mm. Disc: Ø 280 mm.



T4 patch · RUNE-Tag IDX 24 · seed 576808920

Cut along the grey circle. Double tick → circle centre.



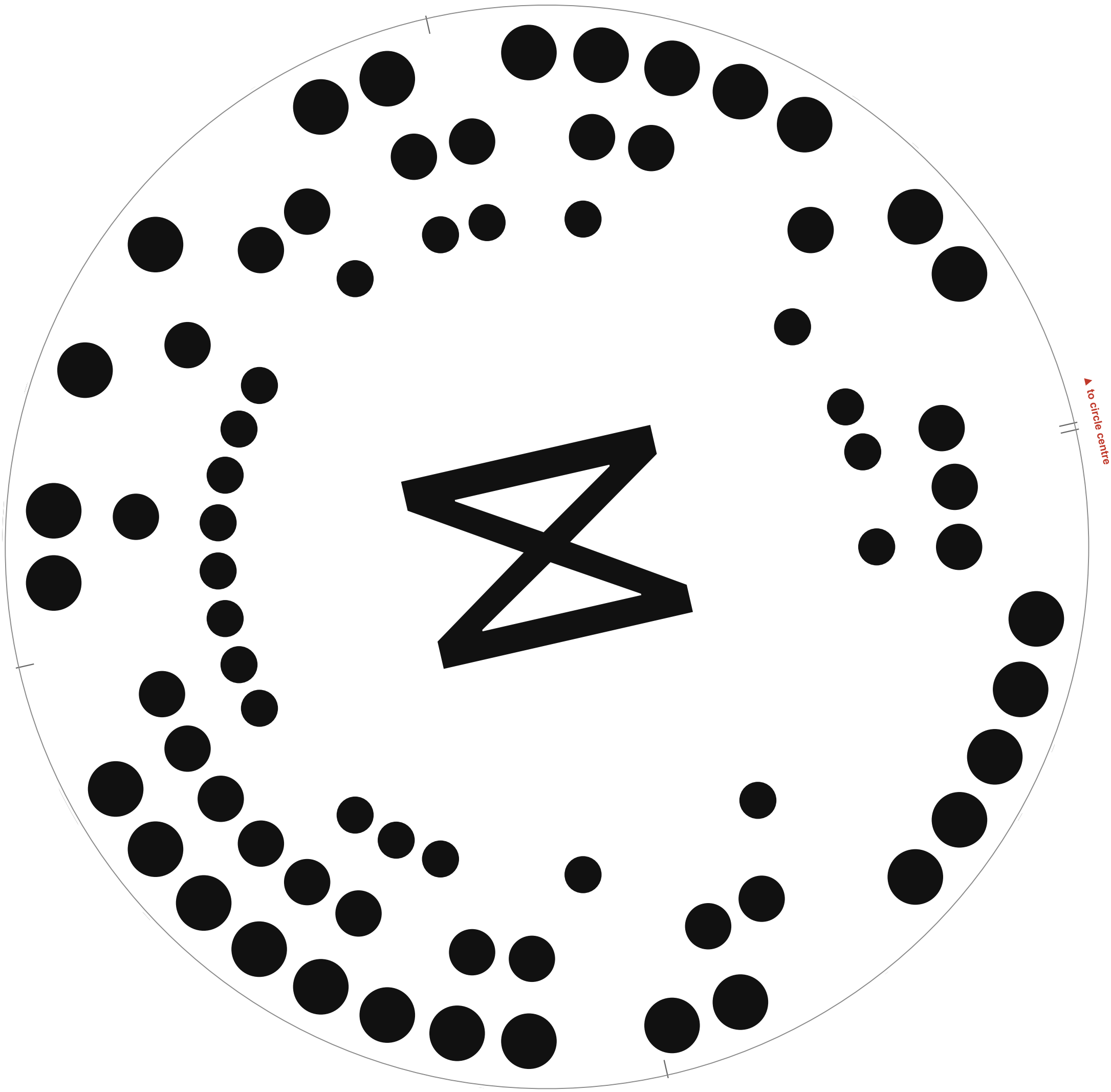


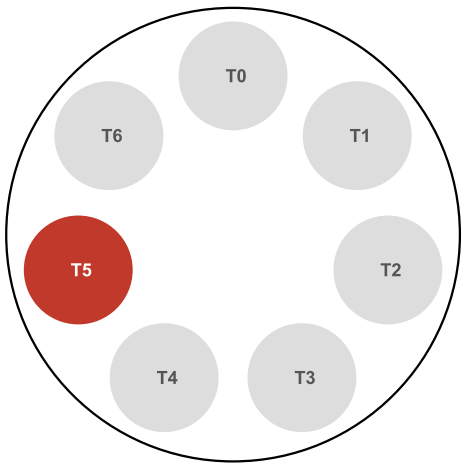
Glue this disc over tag T4 on the assembled poster
(the tag at 6.86 o'clock, lower left).
Line up all four ticks with the ticks on the poster.
Spread glue thinly and evenly; press flat from the centre out.
Dots: outer edge Ø 270 mm. Disc: Ø 280 mm.



T5 patch · RUNE-Tag IDX 26 · seed 576808920

Cut along the grey circle. Double tick → circle centre.



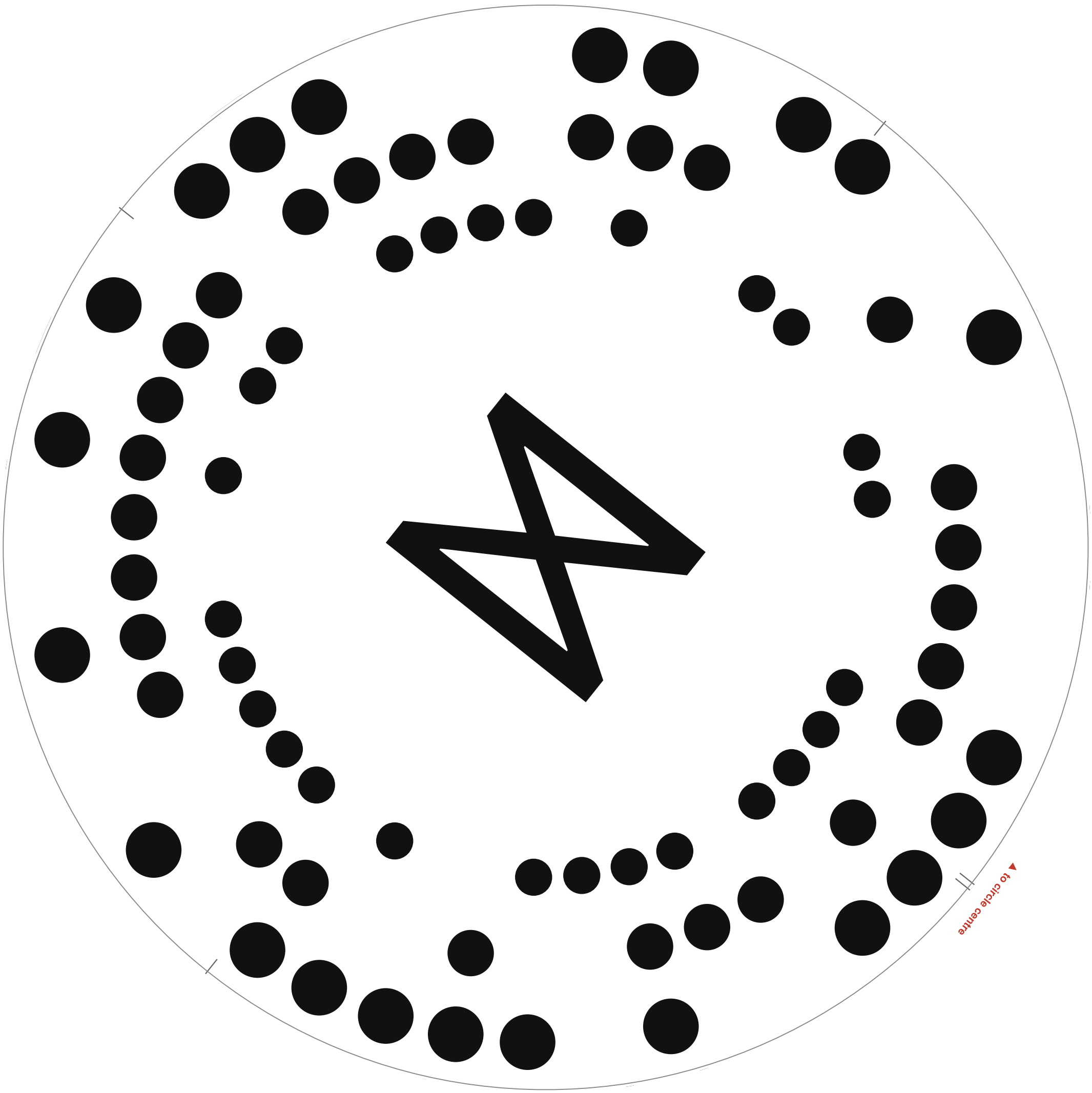


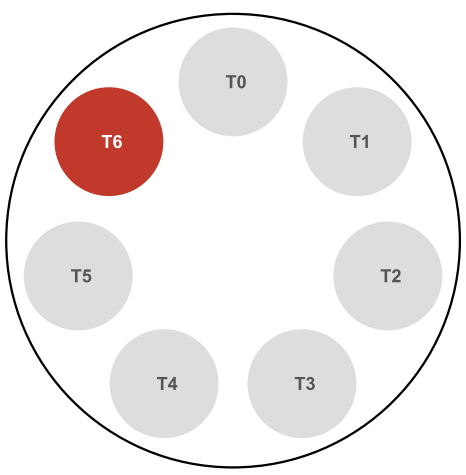
Glue this disc over tag T5 on the assembled poster
(the tag at 8.57 o'clock, left).
Line up all four ticks with the ticks on the poster.
Spread glue thinly and evenly; press flat from the centre out.
Dots: outer edge Ø 270 mm. Disc: Ø 280 mm.



T6 patch · RUNE-Tag IDX 29 · seed 576808920

Cut along the grey circle. Double tick → circle centre.





Glue this disc over tag T6 on the assembled poster
(the tag at 10.29 o'clock, upper left).
Line up all four ticks with the ticks on the poster.
Spread glue thinly and evenly; press flat from the centre out.
Dots: outer edge Ø 270 mm. Disc: Ø 280 mm.

